BT 201 (Semester 4)

Spring 2025

Module: BIOMOLECULES

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MBBT

NUCLEOTIDES

- ✓ A segment of a DNA/RNA molecule comprises four nucleotides: A, T/U, G, and C.

- ✓ **Nucleotides** have three characteristic components:

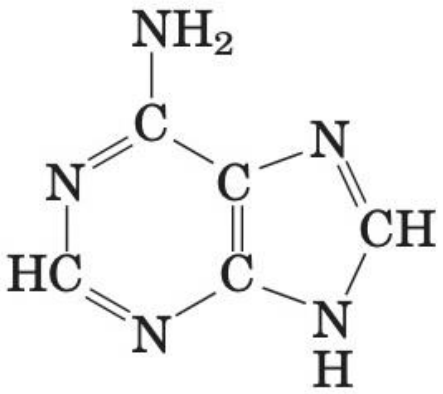
(1) a nitrogenous base,
(2) a pentose, and
(3) one or more phosphates.

The nitrogenous bases are derivatives of two parent compounds, **pyrimidine** and **purine**.

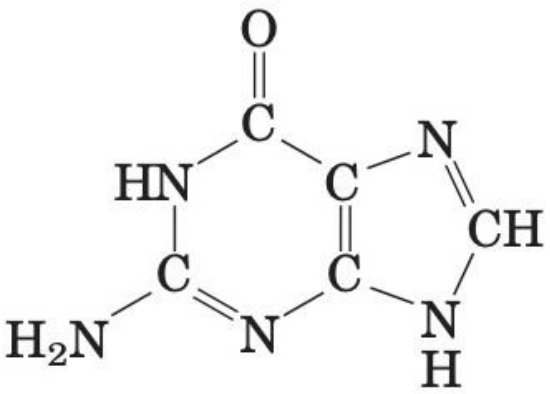
- ✓ DNA and RNA contain two major purine bases, **adenine** (A) and **guanine** (G), and two major pyrimidines. In both DNA and RNA, one of the pyrimidines is **cytosine** (C), but the second common pyrimidine is not the same in both: it is **thymine** (T) in DNA and **uracil** (U) in RNA.

- ✓ Nucleic acids have two kinds of pentoses. The recurring deoxyribonucleotide units of DNA contain 2'-deoxy-D-ribose, and the ribonucleotide units of RNA contain D-ribose. Both types of pentoses are in their *β*-furanose (closed five-membered ring) form. The pentose ring is not planar but occurs in one of a variety of conformations generally described as "puckered."

NUCLEOTIDES

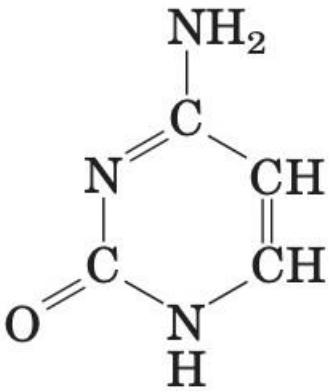


Adenine

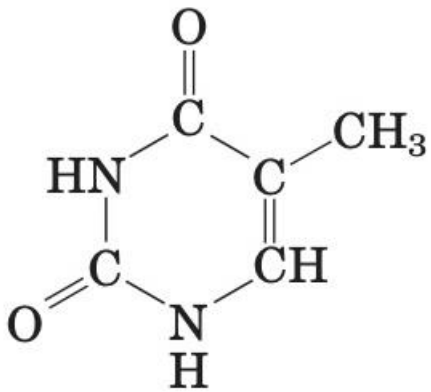


Guanine

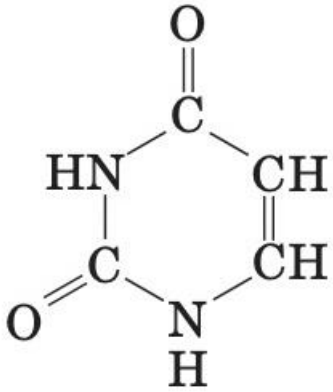
PURINES



Cytosine



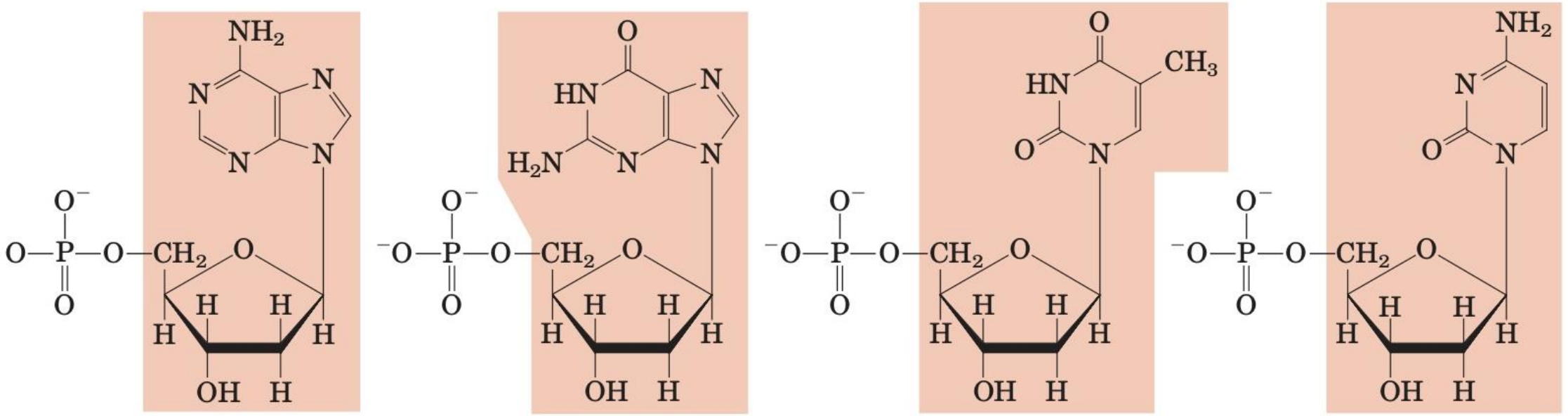
Thymine
(DNA)



Uracil
(RNA)

PYRIMIDINES

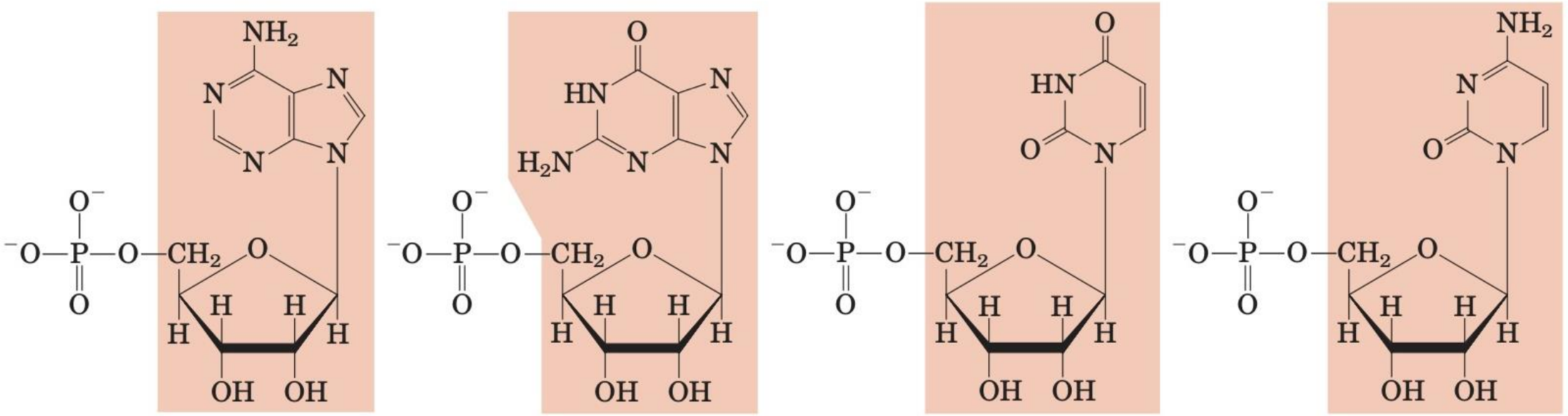
NUCLEOTIDES



Nucleotide:	Deoxyadenylate (deoxyadenosine 5′-monophosphate)	Deoxyguanylate (deoxyguanosine 5′-monophosphate)	Deoxythymidylate (deoxythymidine 5′-monophosphate)	Deoxycytidylate (deoxycytidine 5′-monophosphate)
Symbols:	A, dA, dAMP	G, dG, dGMP	T, dT, dTMP	C, dC, dCMP
Nucleoside:	Deoxyadenosine	Deoxyguanosine	Deoxythymidine	Deoxycytidine

DEOXYRIBONUCLEOTIDES

NUCLEOTIDES



Nucleotide:	Adenylate (adenosine 5′-monophosphate)	Guanylate (guanosine 5′-monophosphate)	Uridylate (uridine 5′-monophosphate)	Cytidylate (cytidine 5′-monophosphate)
Symbols:	A, AMP	G, GMP	U, UMP	C, CMP
Nucleoside:	Adenosine	Guanosine	Uridine	Cytidine

RIBONUCLEOTIDES

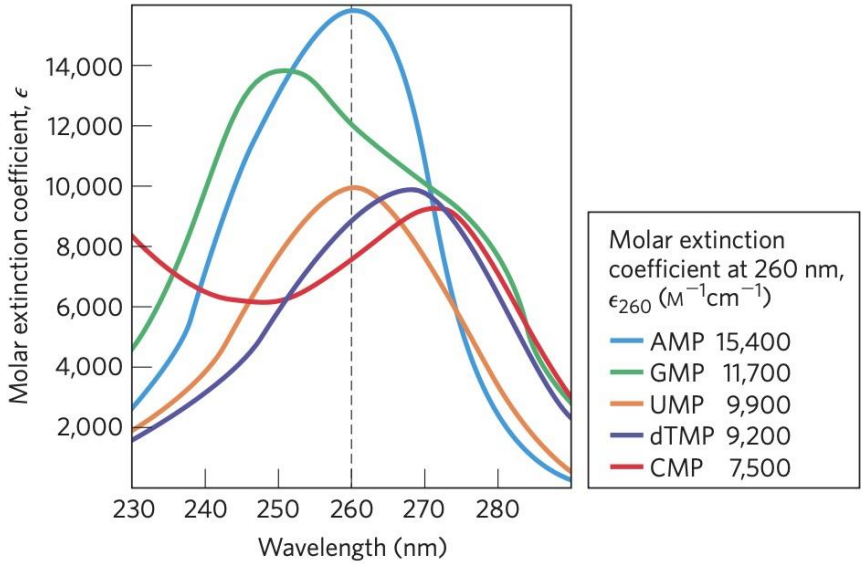
NUCLEOTIDES

- ✓ The purine and pyrimidine bases are hydrophobic and relatively insoluble in water at the near-neutral pH of the cell. At acidic or alkaline pH the bases become charged and their solubility in water increases. Hydrophobic stacking interactions, van der Waals and dipole-dipole interactions are common modes of interactions between the bases.

- ✓ Pyrimidines are planar molecules and purines are very nearly planar, with a slight pucker.

- ✓ Free pyrimidine and purine bases may exist in two or more tautomeric forms depending on the pH. Uracil, for example, occurs in lactam, lactim, and double lactim forms.

- ✓ All nucleotide bases absorb UV light, and nucleic acids are characterized by a strong absorption at wavelengths near 260 nm.



DNA

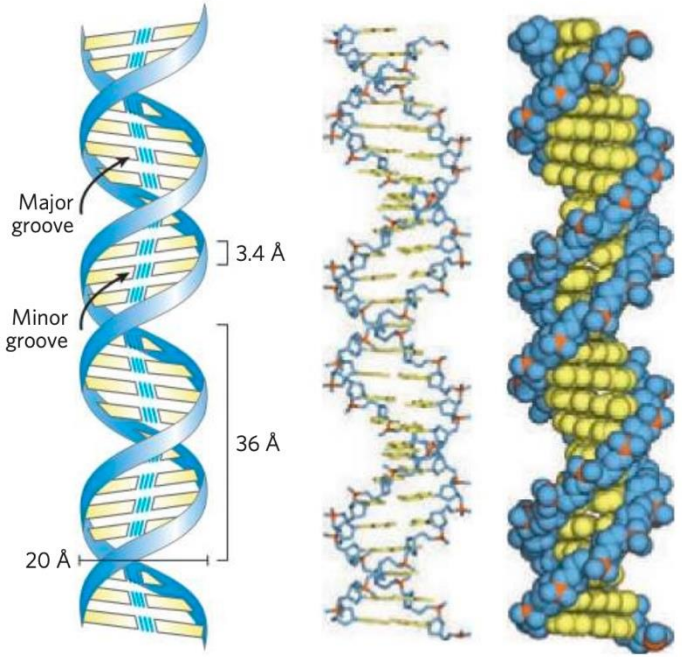
- ✓ DNA was first isolated and characterized by Friedrich Miescher in 1868. He called the phosphorus-containing substance “nuclein.”

- ✓ Not until the 1940s, was there any compelling evidence that DNA was the genetic material. Oswald T. Avery, Colin MacLeod, and Maclyn McCarty found that DNA extracted from a virulent (disease-causing) strain of the bacterium *Streptococcus pneumoniae* when injected into a non-virulent strain of the same bacterium, transformed the non-virulent strain into a virulent strain. Thus, they concluded that DNA from the virulent strain carried the genetic information for virulence.

- ✓ DNA structure was discovered by Watson and Crick in 1953.

James D. Watson

Francis Crick, 1916–2004



DNA structure

DNA

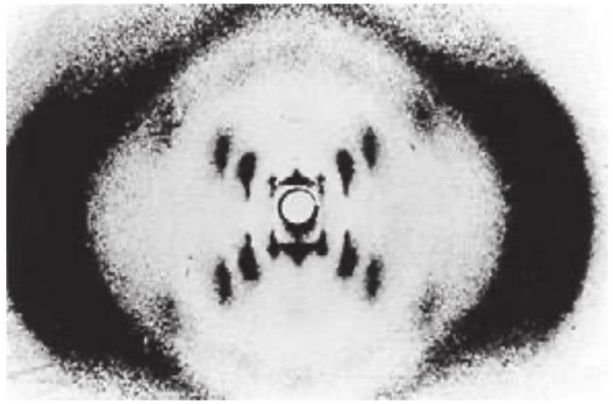
- ✓ Another important clue to the structure of DNA came from the work of Erwin Chargaff and his colleagues, which are collectively known as Chargaff's rules:

1. The base composition of DNA generally varies from one species to another.
2. DNA specimens isolated from different tissues of the same species have the same base composition.
3. The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment.
4. In all cellular DNAs, regardless of the species, the number of adenosine residues is equal to the number of thymidine residues (that is, A = T), and the number of guanosine residues is equal to the number of cytidine residues (G = C). From these relationships it follows that the sum of the purine residues equals the sum of the pyrimidine residues, *i.e.*, A + G = T + C.

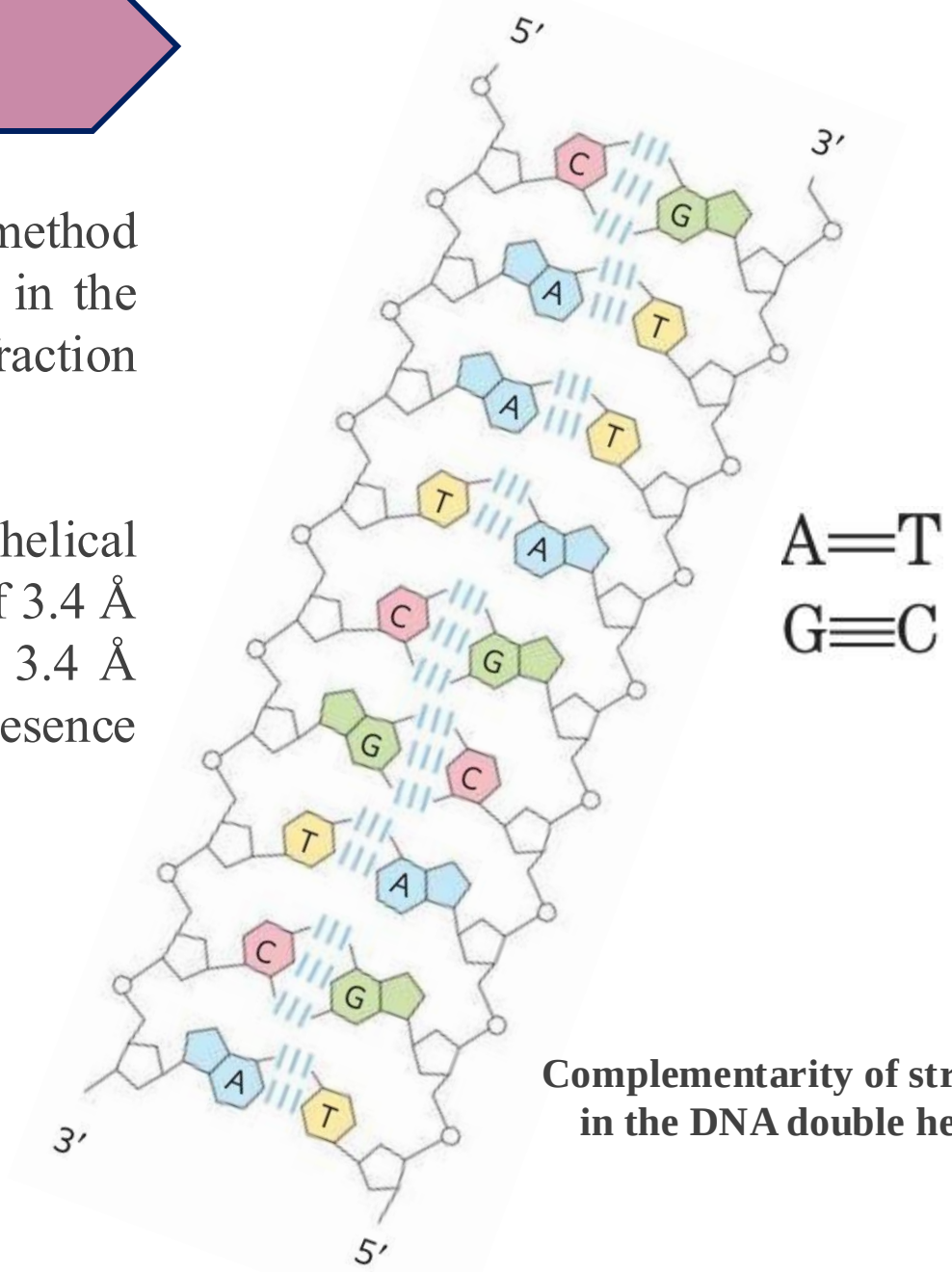
DNA

- ✓ Rosalind Franklin and Maurice Wilkins used the powerful method of X-ray diffraction to analyze DNA fibers. They showed in the early 1950s that DNA produces a characteristic X-ray diffraction pattern

- ✓ From this pattern it was deduced that DNA molecules are helical with two periodicities along their long axis, a primary one of 3.4 Å (vertically stacked bases inside the double helix would be 3.4 Å apart) and a secondary one of 34 Å (accounted for by the presence of 10 base pairs in each complete turn of the double helix).



X-ray diffraction pattern of DNA fibers

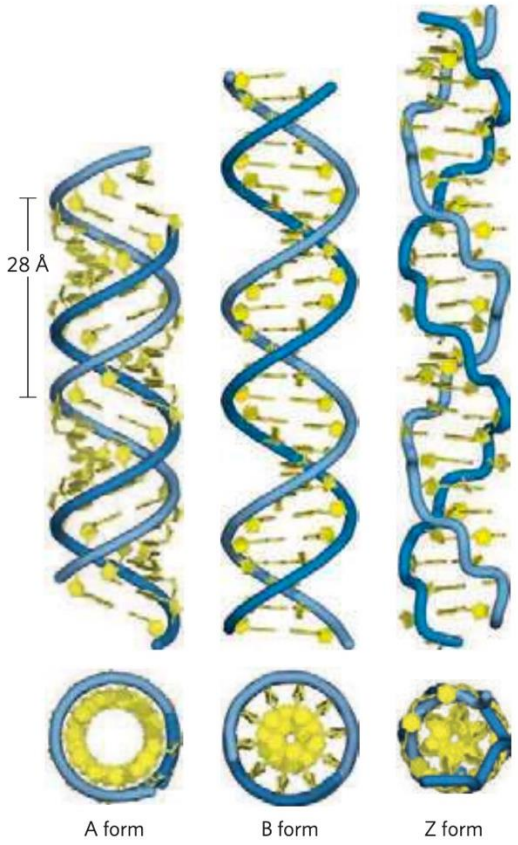


Complementarity of strands in the DNA double helix

DNA

- ✓ The Watson-Crick structure is also referred to as **B-form DNA**, or B-DNA. It is the most stable structure for a random-sequence DNA molecule under physiological conditions.

- ✓ Two structural variants that have been well characterized in crystal structures are the **A** and **Z forms**.



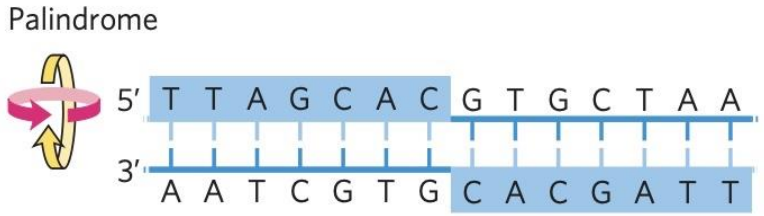
	A form	B form	Z form
Helical sense	Right handed	Right handed	Left handed
Diameter	~26 Å	~20 Å	~18 Å
Base pairs per helical turn	11	10.5	12
Helix rise per base pair	2.6 Å	3.4 Å	3.7 Å
Base tilt normal to the helix axis	20°	6°	7°
Sugar pucker conformation	C-3′ endo	C-2′ endo	C-2′ endo for pyrimidines; C-3′ endo for purines
Glycosyl bond conformation	Anti	Anti	Anti for pyrimidines; syn for purines

Comparison of A, B, and Z forms of DNA

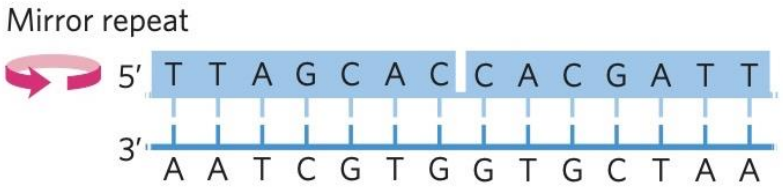
DNA

✓ Certain DNA sequences adopt unusual structures such as:

1. **Palindrome structures:** A palindrome is a word, phrase, or sentence that is spelt identically and read either forward or backward; two examples are ROTATOR and NURSES RUN. The term is applied to regions of DNA with **inverted repeats** of base sequence having twofold symmetry over two strands of DNA.

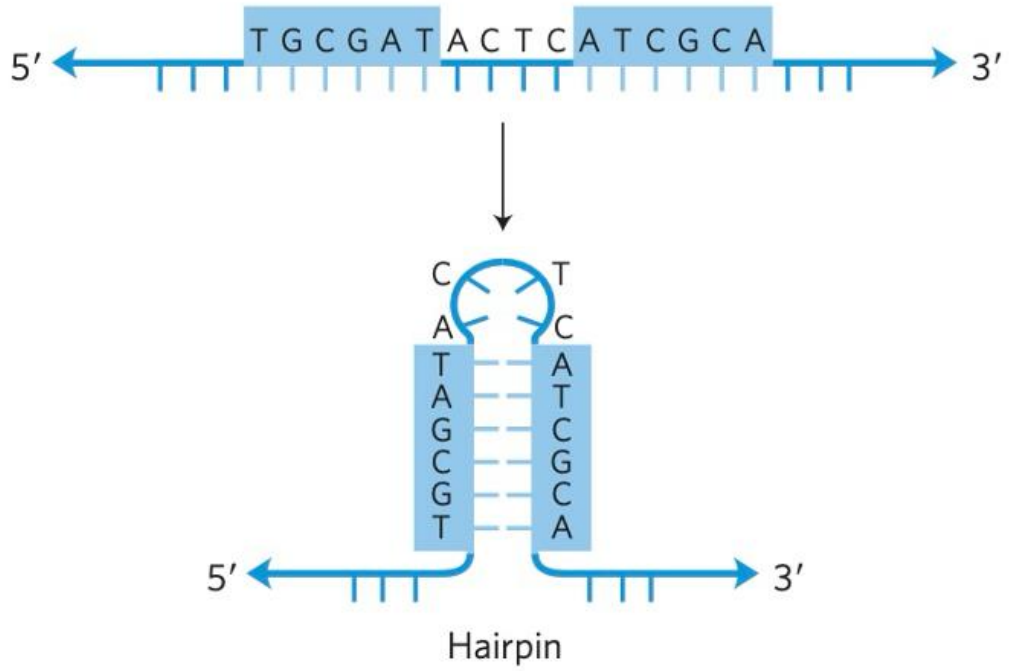


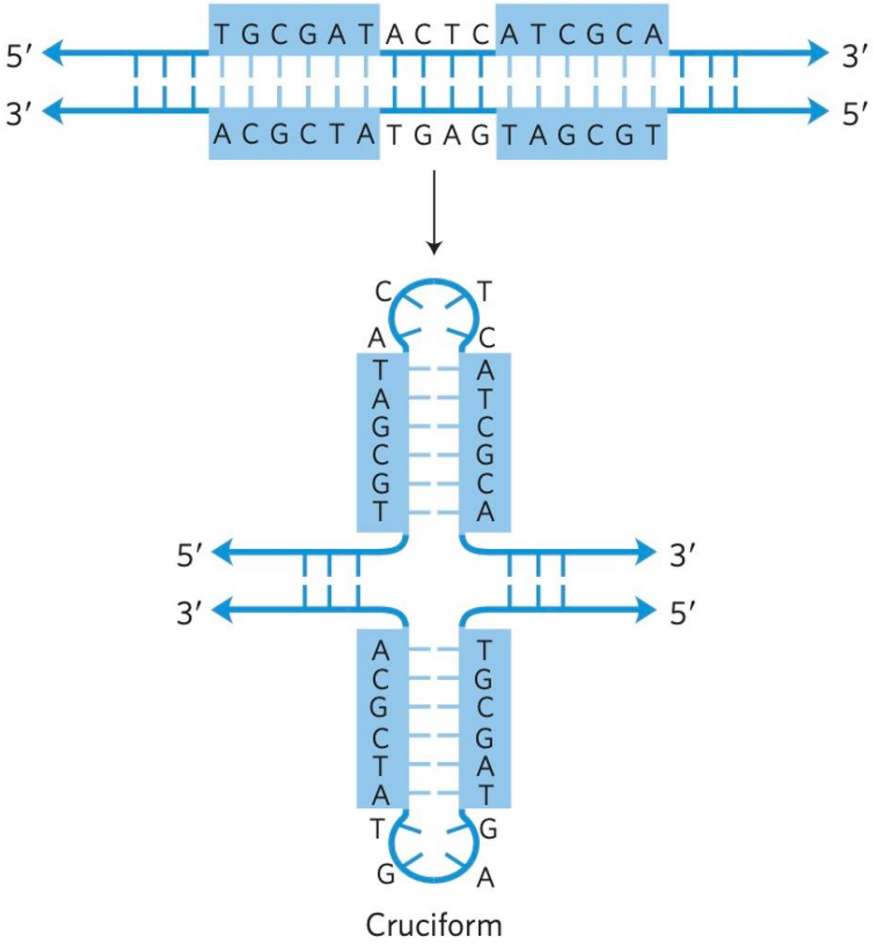
2. **Mirror repeats:** When the inverted repeat occurs within each individual strand of the DNA, the sequence is called a **mirror repeat**. Mirror repeats do not have complementary sequences within the same strand.



DNA

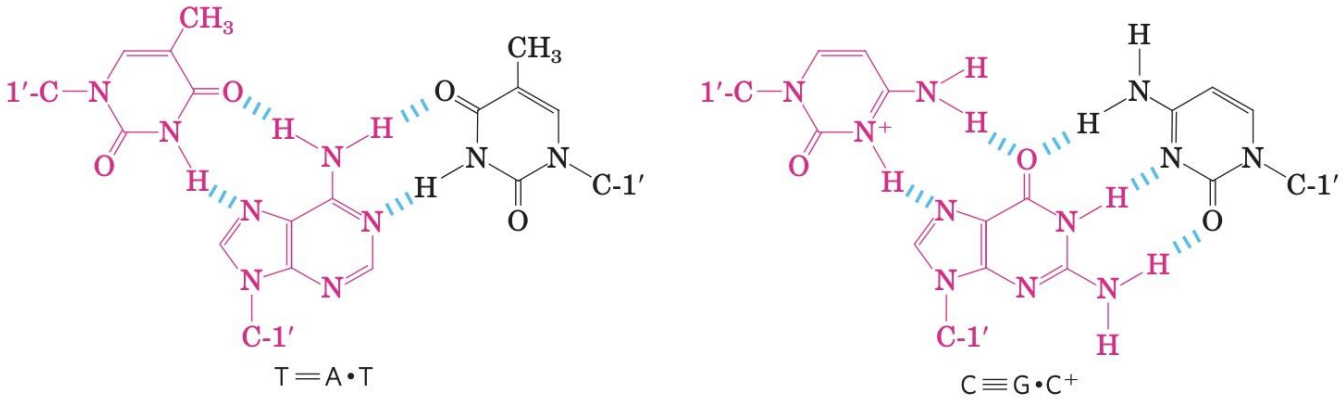
3. Hairpins and Cruciforms: Such sequences are self-complementary within each strand and therefore have the potential to form **hairpin** or **cruciform** (cross-shaped) structures.



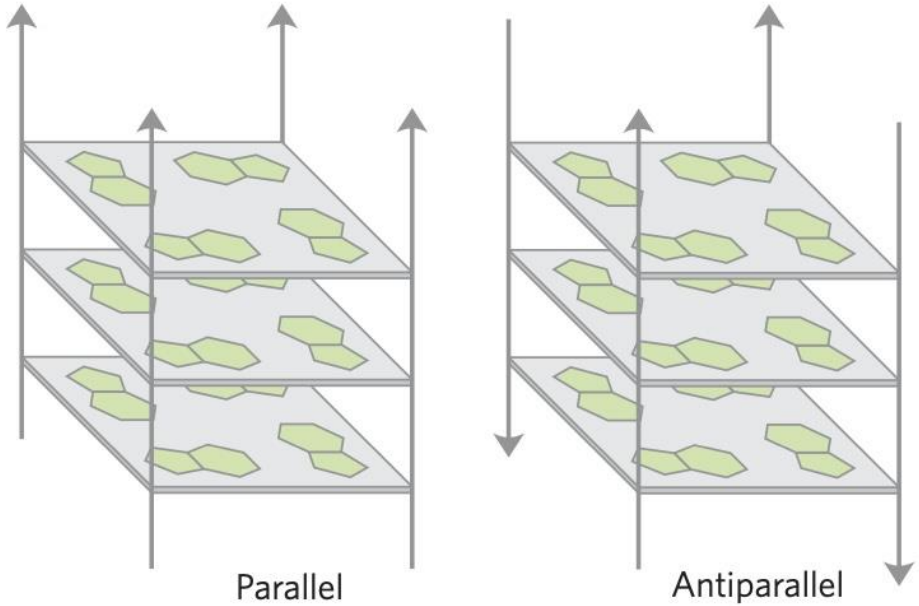


DNA

4. Triplex DNAs and G quadruplex

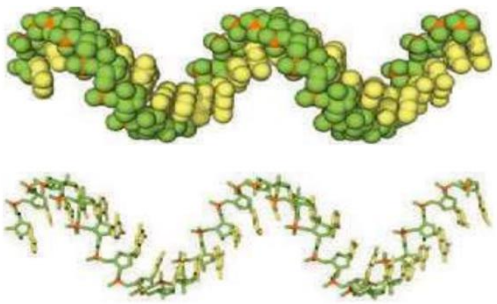


Triplex DNA



G tetraplex

RNA

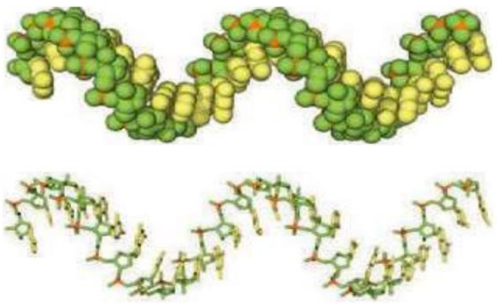


- ✓ The product of transcription of DNA is always single-stranded (ss) RNA.
- ✓ The single strand tends to assume a right-handed helical conformation dominated by base-stacking interactions.
- ✓ RNA can base-pair with complementary regions of either RNA or DNA. Base pairing matches the pattern for DNA: G pairs with C and A pairs with U.
- ✓ The 3D structures of RNAs are complex and unique. Weak interactions, especially base-stacking interactions, help stabilise RNA structures.
- ✓ Various forms of RNA include:

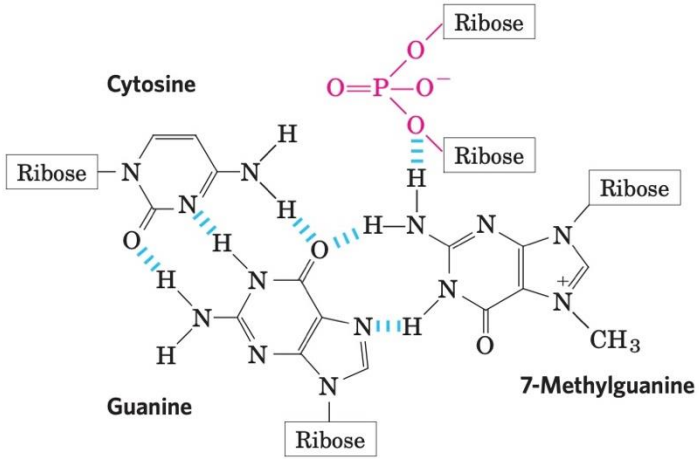
(i) A-form: The predominant ds structure when two strands of RNA with perfectly complementary sequences are paired.

(ii) Z-form: Made in the laboratory under very high-salt or high-temperature conditions.

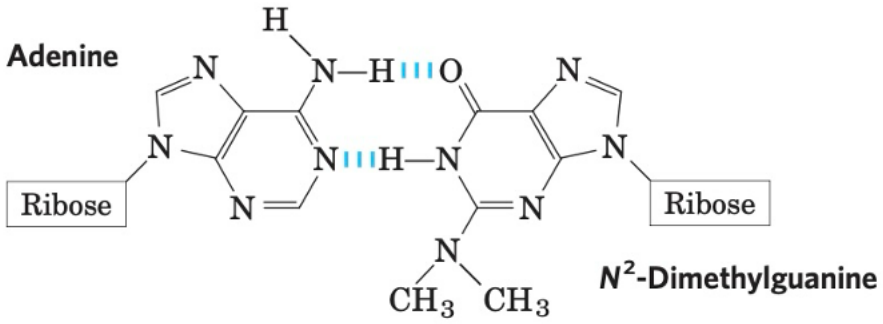
RNA



Some examples of 3D structure in RNA (remember example names):

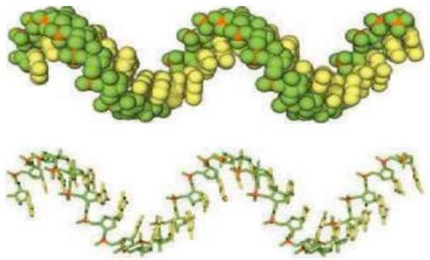


Phenylalanine tRNA of yeast

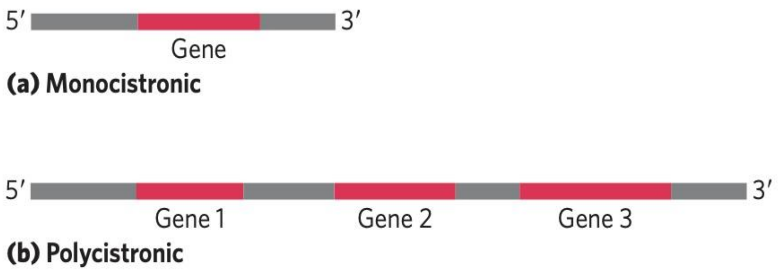


A hammerhead ribozyme derived from certain plant viruses

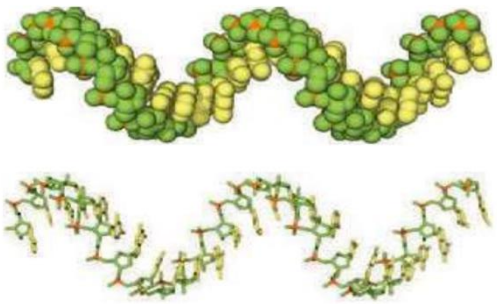
RNA



- ✓ RNA, the second major form of nucleic acid in cells, has many functions.
- ✓ In gene expression, RNA acts as an intermediate by using the information encoded in the DNA to specify the amino acid sequence of a functional protein.
- ✓ RNA is found in both the nucleus and the cytoplasm.
- ✓ In 1961, François Jacob and Jacques Monod proposed the name “messenger RNA” (mRNA) for that portion of the total cellular RNA carrying the genetic information from DNA to the ribosomes (ribosome: a cell structure which synthesises proteins by reading the mRNA and translating it into amino acids). The process of forming mRNA on a DNA template is known as **transcription**.
- ✓ In bacteria and archaea, a single mRNA molecule may code for one or several polypeptide chains. If it carries the code for only one polypeptide, the mRNA is **monocistronic**; if it codes for two or more different polypeptides, the mRNA is **polycistronic**. In eukaryotes, most mRNAs are monocistronic (cistron refers to a gene).



RNA



Different types of RNA:

(i) messenger RNA (mRNA)

(ii) transfer RNA (tRNA)

(iii) ribosomal RNA (rRNA)

(iv) microRNA (miRNA)

(v) long non-coding RNA (lncRNA)

(vi) small nuclear RNA (snRNA)

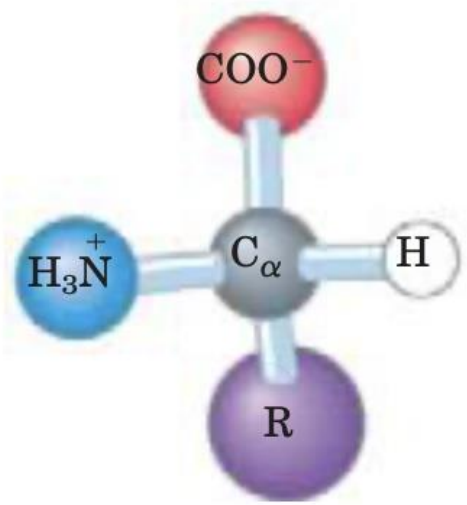
(vii) small nucleolar RNA (snoRNA)

Amino acids

- ✓ Proteins are polymers of amino acids, with each **amino acid residue** joined to its neighbour by a specific type of covalent bond.

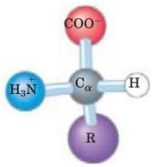
- ✓ 20 different amino acids are commonly found in proteins.

- ✓ All 20 of the common amino acids are α-amino acids. They have a carboxyl group and an amino group bonded to the same carbon atom (the α carbon). They differ from each other in their side chains, or **R groups**, which vary in structure, size, and electric charge, and which influence the solubility of the amino acids in water.



General structure of an amino acid: This structure is common to all but one of the -amino acids. (Proline, a cyclic amino acid, is the exception.) The R group, or side chain (purple), attached to the α carbon (grey) is different in each amino acid.

Amino acids



Nonpolar, aliphatic R groups		
Glycine	Gly	G
Alanine	Ala	A
Proline	Pro	P
Valine	Val	V
Leucine	Leu	L
Isoleucine	Ile	I
Methionine	Met	M

Aromatic R groups		
Phenylalanine	Phe	F
Tyrosine	Tyr	Y
Tryptophan	Trp	W

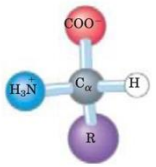
Polar, uncharged R groups		
Serine	Ser	S
Threonine	Thr	T
Cysteine¶	Cys	C
Asparagine	Asn	N
Glutamine	Gln	Q

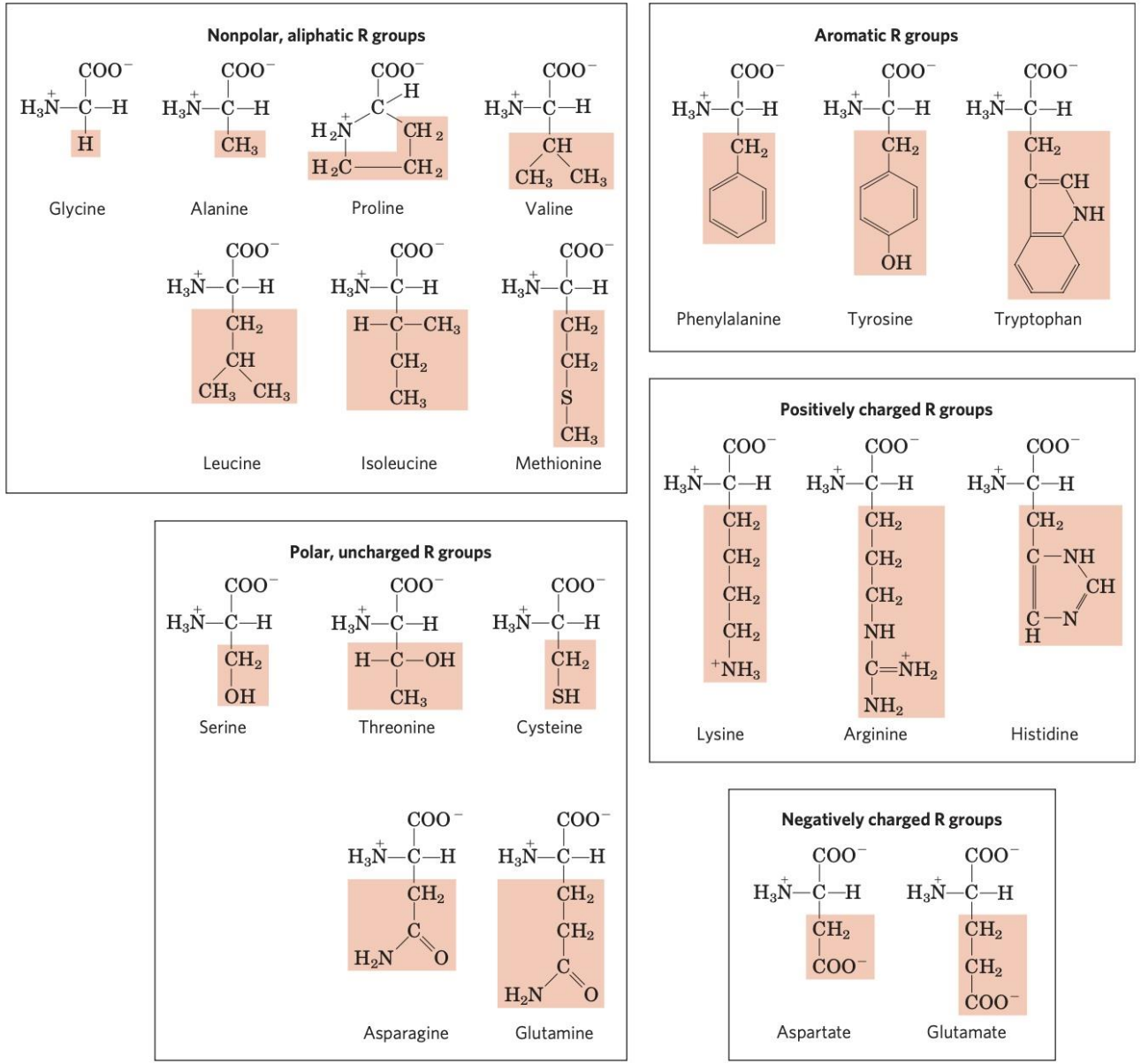
Positively charged R groups		
Lysine	Lys	K
Histidine	His	H
Arginine	Arg	R

Negatively charged R groups		
Aspartate	Asp	D
Glutamate	Glu	E

Remember any two amino acids along with their abbreviations from each group.

Amino acids

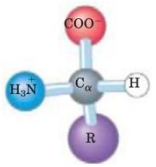




The 20 common amino acids of proteins: The structural formulas show the state of ionization that would predominate at pH 7.0. The unshaded portions are those common to all the amino acids; the shaded portions are the R groups.

Serine is the only amino acid containing S.

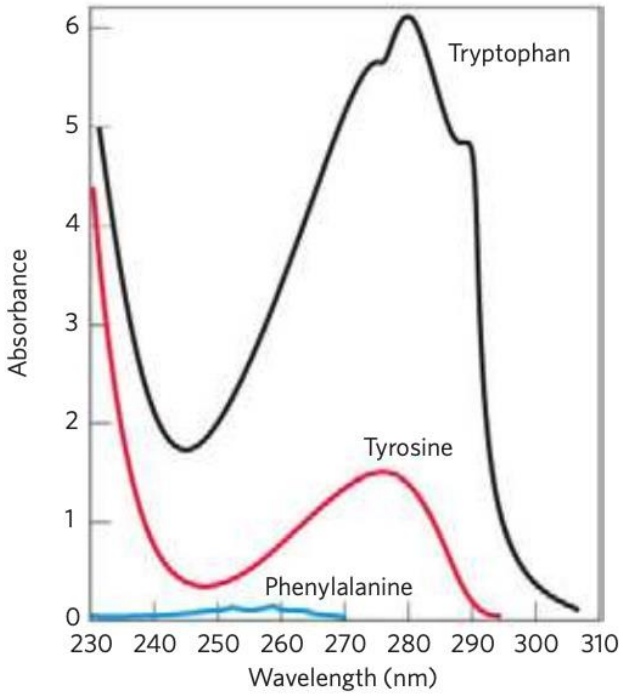
Amino acids



- ✓ Tryptophan and tyrosine, and to a much lesser extent phenylalanine, absorb ultraviolet light. This accounts for the characteristic strong absorbance of light by most proteins at a wavelength of 280 nm, a property exploited by researchers in the characterization of proteins.

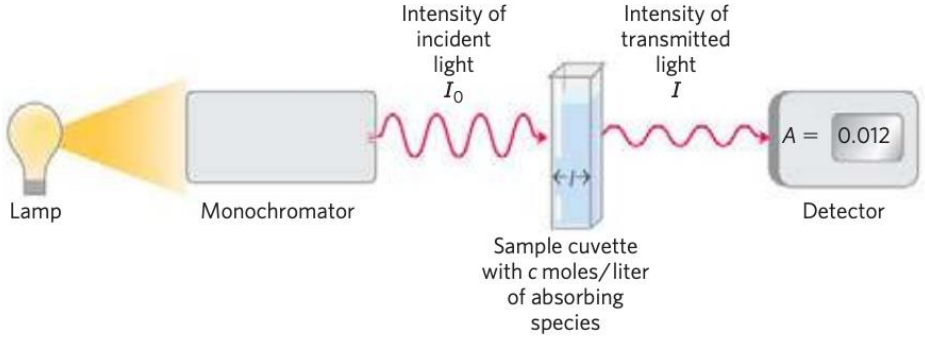
- ✓ Measurement of light by a spectrophotometer is used to detect and identify molecules and to measure their concentration in solution.

- ✓ The fraction of the incident light absorbed by a solution at a given wavelength is related to the thickness of the absorbing layer (path length) and the concentration of the absorbing species. These two relationships are combined into the Lambert-Beer law,



$$\log \frac{I_0}{I} = \varepsilon c l$$

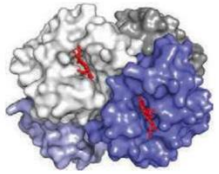
I_0 = Intensity of the incident light,
I = Intensity of the transmitted light
I/I_0 = Transmittance,
ε = Molar extinction coefficient (in units of litres per mole-cm)
c = Concentration of the absorbing species (in moles L^{-1})
l = path length of the light-absorbing sample (in cm)



The principal components of a spectrophotometer

Remember the equation along with what each symbol denotes.

Proteins

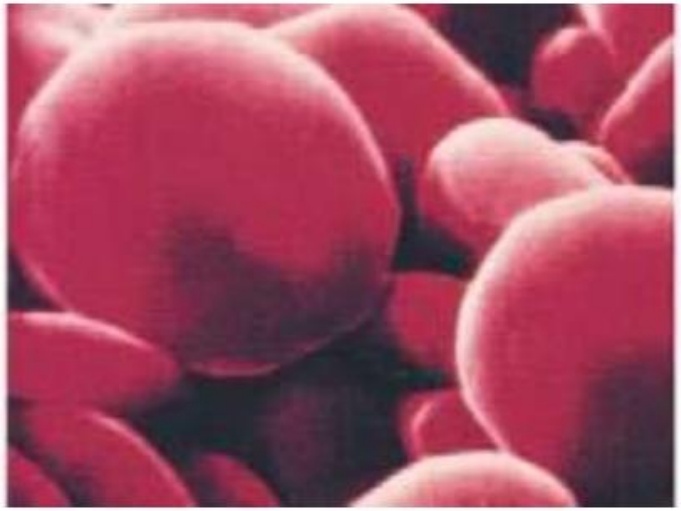


- ✓ We now turn to polymers of amino acids, **peptides** and **proteins**.
- ✓ Biologically occurring polypeptides range in size from small to very large, consisting of two or three to thousands of linked amino acid residues.

(a)

The light produced by fireflies is the result of a reaction involving the protein luciferin and ATP, catalyzed by the enzyme luciferase.

(b)

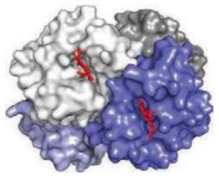


Erythrocytes contain large amounts of the oxygen-transporting protein haemoglobin.

(c)

The protein keratin, formed by all vertebrates, is the chief structural component of hair, scales, horn, wool, nails, and feathers.

Proteins



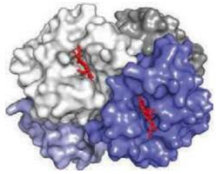
- ✓ Peptides are chains of amino acids: Two amino acid molecules can be covalently joined through a substituted amide linkage, termed a **peptide bond**, to yield a dipeptide. Such a linkage is formed by removal of the elements of water (dehydration) from the α-carboxyl group of one amino acid and the α-amino group of another.

- ✓ Three amino acids can be joined by two peptide bonds to form a tripeptide; similarly, four amino acids can be linked to form a tetrapeptide, five to form a pentapeptide, and so forth. When a few amino acids are joined in this fashion, the structure is called an **oligopeptide**. When many amino acids are joined, the product is called a **polypeptide**.

- ✓ Peptide bond formation is an example of a condensation reaction, a common class of reactions in living cells.

- ✓ Although the terms “protein” and “polypeptide” are sometimes used interchangeably, molecules referred to as polypeptides generally have molecular weights below 10,000 Da, and those called proteins have higher molecular weights.

Proteins

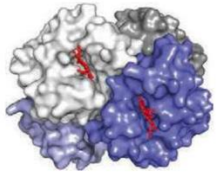


Some proteins consist of a single polypeptide chain, but others called **multi-subunit** proteins, have two or more polypeptides associated noncovalently. Haemoglobin, for example, has four polypeptide subunits: two identical α chains and two identical β chains, all four held together by noncovalent interactions.

	Molecular weight	Number of residues	Number of polypeptide chains
Cytochrome *c* (human)	12,400	104	1
Ribonuclease A (bovine pancreas)	13,700	124	1
Lysozyme (chicken egg white)	14,300	129	1
Myoglobin (equine heart)	16,700	153	1
Chymotrypsin (bovine pancreas)	25,200	241	3
Chymotrypsinogen (bovine)	25,700	245	1
Hemoglobin (human)	64,500	574	4
Serum albumin (human)	66,000	609	1
Hexokinase (yeast)	107,900	972	2
RNA polymerase (*E. coli*)	450,000	4,158	5
Apolipoprotein B (human)	513,000	4,536	1
Glutamine synthetase (*E. coli*)	619,000	5,628	12

Remember the names of any four proteins along with the organism's name.

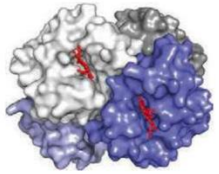
Proteins



- ✓ The amino acid composition of proteins is also highly variable.
- ✓ The 20 common amino acids rarely occur in equal amounts in a protein. Some amino acids may occur only once or not at all in a given type of protein; others may occur in large numbers.
- ✓ The average molecular weight of an amino acid residue in a protein is about 110.

Amino acid	Bovine cytochrome *c*	
	Number of residues per molecule	Percentage of total*
Ala	6	6
Arg	2	2
Asn	5	5
Asp	3	3
Cys	2	2
Gln	3	3
Glu	9	9
Gly	14	13
His	3	3
Ile	6	6
Leu	6	6
Lys	18	17
Met	2	2
Phe	4	4
Pro	4	4
Ser	1	1
Thr	8	8
Trp	1	1
Tyr	4	4
Val	3	3
Total	104	102

Proteins

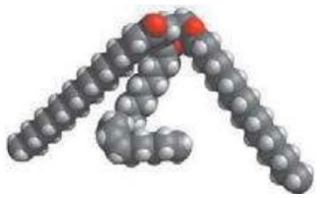


- ✓ A typical protein usually has one or more stable three-dimensional structures, or conformations, that reflect its function. Some proteins have segments that are intrinsically disordered.

- ✓ Secondary structure is the local spatial arrangement of the main-chain atoms in a selected segment of a polypeptide chain. The most common regular secondary structures are the α helix, the β conformation, and random coils. Circular dichroism (CD) spectroscopy is a method for assessing common secondary structure and monitoring folding in proteins.

- ✓ The tertiary structure is the complete 3D structure of a polypeptide chain. Many proteins fall into one of two general classes of proteins based on tertiary structure: fibrous and globular. Fibrous proteins, which serve mainly structural roles, have simple repeating elements of secondary structure (α-Keratin of hair, feathers and nails, collagen of tendons and bone matrix, silk fibroin). Globular proteins have more complicated tertiary structures, often containing several types of secondary structures in the same polypeptide chain. The first globular protein structure to be determined, by X-ray diffraction methods, was that of myoglobin.

Lipids

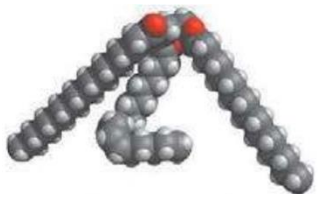


- ✓ Biological lipids are a chemically diverse group of compounds, the defining feature of which is their insolubility in water.

- ✓ The biological functions of the lipids are diverse:

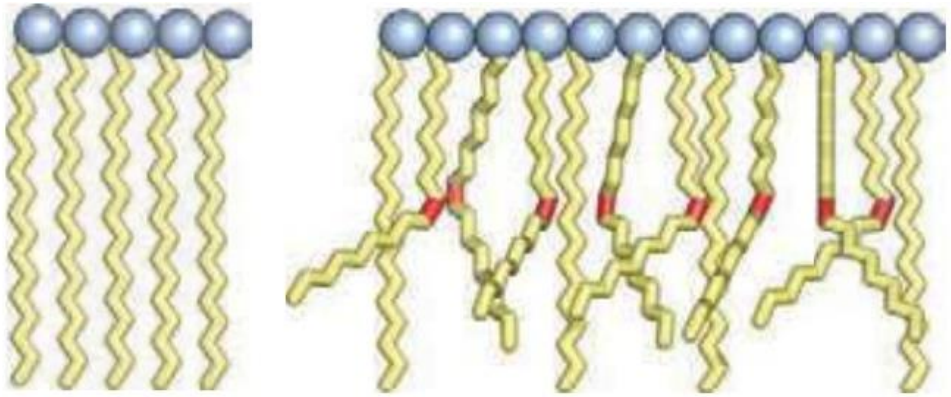
(i) Fats and oils are the principal stored forms of energy in many organisms.
(ii) Phospholipids and sterols are major structural elements of biological membranes.
(iii) Other lipids, although present in relatively small quantities, play crucial roles as enzyme cofactors, electron carriers, light-absorbing pigments, hydrophobic anchors for proteins, “chaperones” to help membrane proteins fold, emulsifying agents in the digestive tract, hormones, and intracellular messengers.

Lipids



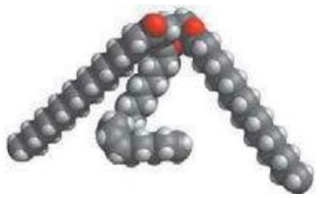
- ✓ The fats and oils used almost universally as stored forms of energy in living organisms are derivatives of **fatty acids**.

- ✓ Fatty acids are carboxylic acids with hydrocarbon chains ranging from 4 to 36 carbons long (C_4 to C_{36}). In some fatty acids, this chain is unbranched and fully saturated (contains no double bonds); in others, the chain contains one or more double bonds.



Saturated FAs Saturated FAs + Unsaturated FAs

Lipids

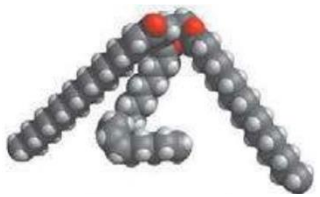


- ✓ The family of **polyunsaturated fatty acids (PUFAs)** with a double bond between the third and fourth carbon from the methyl end of the chain are of special importance in human nutrition.

- ✓ PUFAs with a double bond between C-3 and C-4 are called **omega-3 (ω-3) fatty acids**, and those with a double bond between C-6 and C-7 are **omega-6 (ω-6) fatty acids.** An imbalance of ω-6 and ω-3 PUFAs in the diet is associated with an increased risk of cardiovascular disease. The optimal dietary ratio of ω-6 to ω-3 PUFAs is between 1:1 and 4:1. The "Mediterranean diet," which has been associated with lowered cardiovascular risk, is richer in ω-3 PUFAs, obtained in leafy vegetables (salads) and fish oils.

- ✓ Fish oil supplements which are especially rich in eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) are often prescribed for individuals with a history of cardiovascular disease.

Lipids



- ✓ The simplest lipids constructed from fatty acids are the **triacylglycerols**, also referred to as triglycerides, fats, or neutral fats.

- ✓ Triacylglycerols are composed of three fatty acids each in ester linkage with a single glycerol.

- ✓ In vertebrates, specialized cells called adipocytes, or fat cells, store large amounts of triacylglycerols as fat droplets that nearly fill the cell. Triacylglycerols are also stored as oils in the seeds of many types of plants, providing energy and biosynthetic precursors during seed germination.

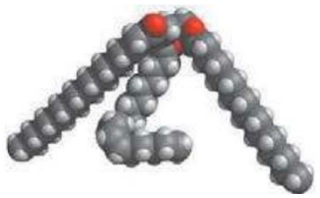
- ✓ Adipocytes and germinating seeds contain **lipases**, enzymes that catalyze the hydrolysis of stored triacylglycerols, releasing fatty acids for export to sites where they are required as fuel.

- ✓ Two significant advantages to using triacylglycerols as stored fuels rather than polysaccharides (glycogen/ starch):

(i) The C atoms of fatty acids are more reduced than those of sugars, and oxidation of triacylglycerols yields more than twice as much energy, gram for gram, as the oxidation of carbohydrates.

(ii) Triacylglycerols are hydrophobic and thus not hydrated, the organism that carries fat as fuel does not have to carry the extra weight of water of hydration that is associated with stored polysaccharides (2 g per gram of polysaccharide).

Lipids



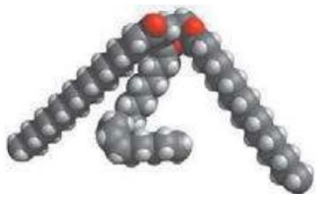
✓ Structural lipids in membranes:

The central architectural feature of biological membranes is a double layer of lipids, which acts as a barrier to the passage of polar molecules and ions. Membrane lipids are amphipathic: one end of the molecule is hydrophobic, the other hydrophilic. Their hydrophobic interactions with each other and their hydrophilic interactions with water direct their packing into sheets called membrane bilayers.

(i) The polar lipids, with polar heads and nonpolar tails, are major components of membranes. The most abundant are the glycerophospholipids, which contain fatty acids esterified to two of the hydroxyl groups of glycerol, and a second alcohol, the head group, esterified to the third hydroxyl of glycerol via a phosphodiester bond. Other polar lipids are the sterols.

(ii) Glycerophospholipids differ in the structure of their head group; common glycerophospholipids are phosphatidylethanolamine and phosphatidylcholine. The polar heads of the glycerophospholipids are charged at pH near 7.

Lipids



(iii) Sphingolipids contain sphingosine, a long- chain aliphatic amino alcohol, but no glycerol. Sphingomyelin has, in addition to phosphoric acid and choline, two long hydrocarbon chains, one contributed by a fatty acid and the other by sphingosine. Three other classes of sphingolipids are cerebrosides, globosides, and gangliosides, which contain sugar components.

(iv) Sterols have four fused rings and a hydroxyl group. Cholesterol, the major sterol in animals, is both a structural component of membranes and precursor to a wide variety of steroids.

(v) Chloroplast membranes are rich in galactolipids, composed of a diacylglycerol with one or two linked galactose residues, and sulfolipids, diacylglycerols with a linked sulfonated sugar residue and thus a negatively charged head group.

(vi) Some archaea have unique membrane lipids, with long-chain alkyl groups ether-linked to glycerol at both ends and sugar residues and/or phosphate joined to the glycerol to provide a polar or charged head group. These lipids are stable under harsh conditions in which these archaea live.

Sugars, Starch & Cellulose

- ✓ During active photosynthesis in bright light, a plant leaf produces more carbohydrate than it needs for generating energy or synthesizing precursors. The excess is converted to sucrose and transported to other parts of the plant, to be used as fuel or stored.

- ✓ In most plants, starch is the main storage form, but in a few plants, such as sugar beet and sugar- cane, sucrose is the primary storage form.

- ✓ The synthesis of sucrose and starch occurs in different cellular compartments (cytosol and plastids, respectively), and these processes are coordinated by a variety of regulatory mechanisms that respond to changes in light level and photosynthetic rate.

- ✓ The synthesis of sucrose and starch is important to the plant but also to humans: starch provides more than 80% of human dietary calories worldwide.

Sugars, Starch & Cellulose

- ✓ Starch is a high molecular weight polymer of D-glucose in (α1 ⟶ 4) linkage. It is synthesized in chloroplasts for temporary storage as one of the stable end products of photosynthesis, and for long-term storage it is synthesized in the amyloplasts of the non-photosynthetic parts of plants—seeds, roots, and tubers (underground stems).

$$\text{Starch}_n + \text{glucose 1-phosphate} + \text{ATP} \longrightarrow \text{starch}_{n+1} + \text{ADP} + 2\text{P}_\text{i}$$

$$\Delta G'^\circ = -50 \text{ kJ/mol}$$

- ✓ Many types of bacteria store carbohydrate in the form of glycogen (highly branched starch). Bacteria, like plant plastids, use ADP-glucose as the activated form of glucose, whereas animal cells use UDP-glucose.

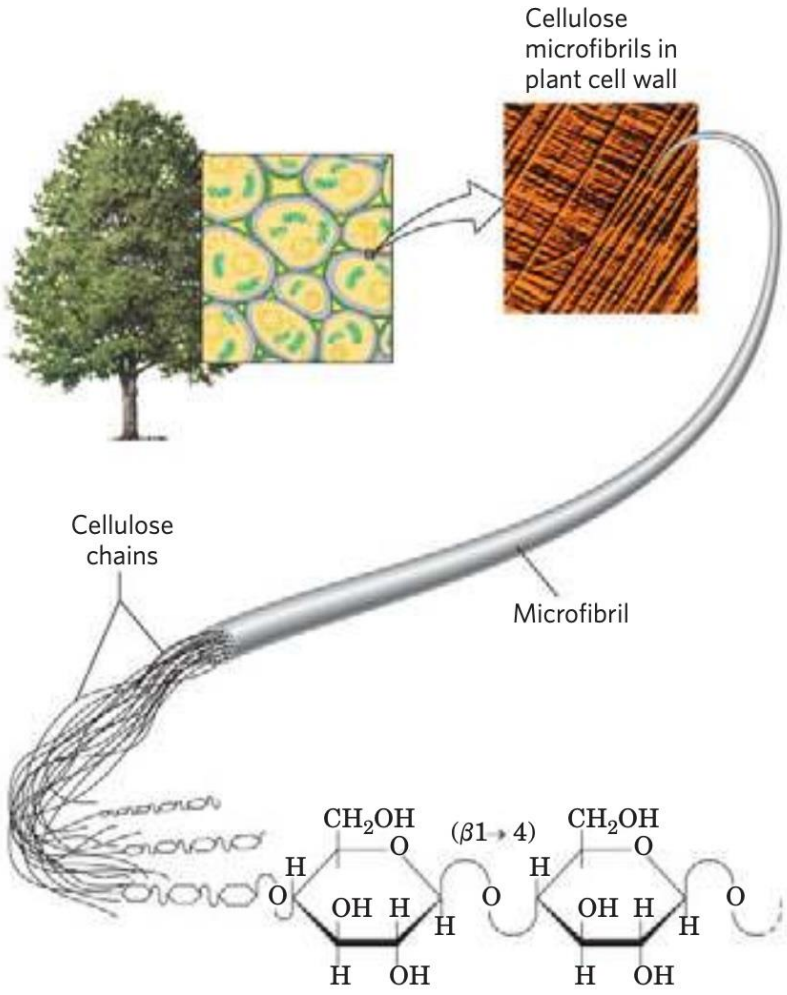
- ✓ ADP-glucose is the substrate for starch synthesis in plant plastids and for glycogen synthesis in bacteria. Enzymes involved: inorganic pyrophosphatase, starch synthase.

- ✓ UDP-glucose is the substrate for sucrose synthesis in the cytosol of leaf cells. Enzymes involved: aldolase, fructose 1,6-bisphosphatase, sucrose 6-phosphate synthase, sucrose 6-phosphate phosphatase.

Remember the equation and enzyme names.

Sugars, Starch & Cellulose

- ✓ Cellulose is a major constituent of plant cell walls, providing strength and rigidity and preventing the swelling of the cell and rupture of the plasma membrane that might result when osmotic conditions favour water entry into the cell.

- ✓ The structure of cellulose is simple: linear polymers of thousands of (β1⟶4)-linked D-glucose units, assembled into bundles of about 36 chains, which aggregate side by side to form a microfibril.

- ✓ Cellulose synthesis takes place in terminal complexes (rosettes) in the plasma membrane. Each cellulose chain begins as a sitosterol dextrin formed inside the cell. It then flips to the outside, where the oligosaccharide portion is transferred to cellulose synthase in the rosette and is then extended. Each rosette produces 36 separate cellulose chains simultaneously and in parallel. The chains crystallize into one of the microfibrils that form the cell wall.